

# Sensory Health and Measurement

Vision, hearing, vestibular, smell, taste, perception, assistive technology, and sensor engineering

Volume 03K of the Health Atlas Research Suite

## Purpose

Integrate peripheral sensory organs, neural coding, perception, development, function and assistive technology into a unified measurement and intervention framework.

**Evidence, guidance, and standards cutoff: 13 August 2026**

August 2026

# Document status and intended use

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This is a technical evidence synthesis and research-engineering reference, not individualized medical advice, a substitute for clinical evaluation, or authorization to use an experimental device on people. It distinguishes established knowledge from uncertain inference and frontier research. Diagnostic and therapeutic decisions require qualified professionals, appropriate reference standards, and applicable ethics and regulatory review.

## Suite relationship

This volume is read with the Atlas of Human Health (master taxonomy), the Library Guide and Crosswalk, and the other domain volumes identified in the cross-reference chapter. Volume 03A, Cardiovascular Health and Measurement Engineering, remains the reference model for the suite.

## Evidence boundary

This volume does not replace urgent evaluation of sudden sensory loss, prescribe hearing/vision devices or diagnose central/perceptual disorders from consumer tests. Sensory measurements are stimulus-, protocol-, language- and context-dependent.

# How to use this volume

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Sensory health is the capacity to detect, discriminate and interpret relevant information well enough to communicate, navigate, learn, work and participate. It cannot be reduced to a threshold: acuity, contrast, speech-in-noise, adaptation, multisensory integration, fatigue, environment and personal goals all matter. Device outputs require calibration to physical stimuli and validation against real-world function.

## Questions this volume is designed to answer

1. How should organ function, neural processing and participation be separated and connected?
2. Which tests measure thresholds versus real-world performance?
3. How do development, aging and multisensory compensation alter interpretation?
4. How should consumer screens and assistive devices be validated?
5. How can systems distinguish impairment from disability created by inaccessible environments?

## Scope

- Visual, auditory, vestibular, olfactory and gustatory physiology and perception.
- Common developmental, degenerative, vascular, infectious, traumatic and neural disorders.
- Psychophysical, optical, acoustic, electrophysiologic, motion and imaging measurement.
- Medical/surgical treatment, rehabilitation and assistive/access technologies.
- Consumer screening and AI validation.

## Reading conventions

- **Established:** replicated mechanisms, mature measurement standards, or interventions supported by convergent evidence.
- **Context-limited:** useful under defined populations, settings, reference methods, or intended uses; transportability must be demonstrated.
- **Frontier:** biologically plausible or technically promising but not yet validated for broad clinical claims.
- **Measurement claim:** always interpreted as an inference from a signal chain rather than a direct view of the latent biological state.

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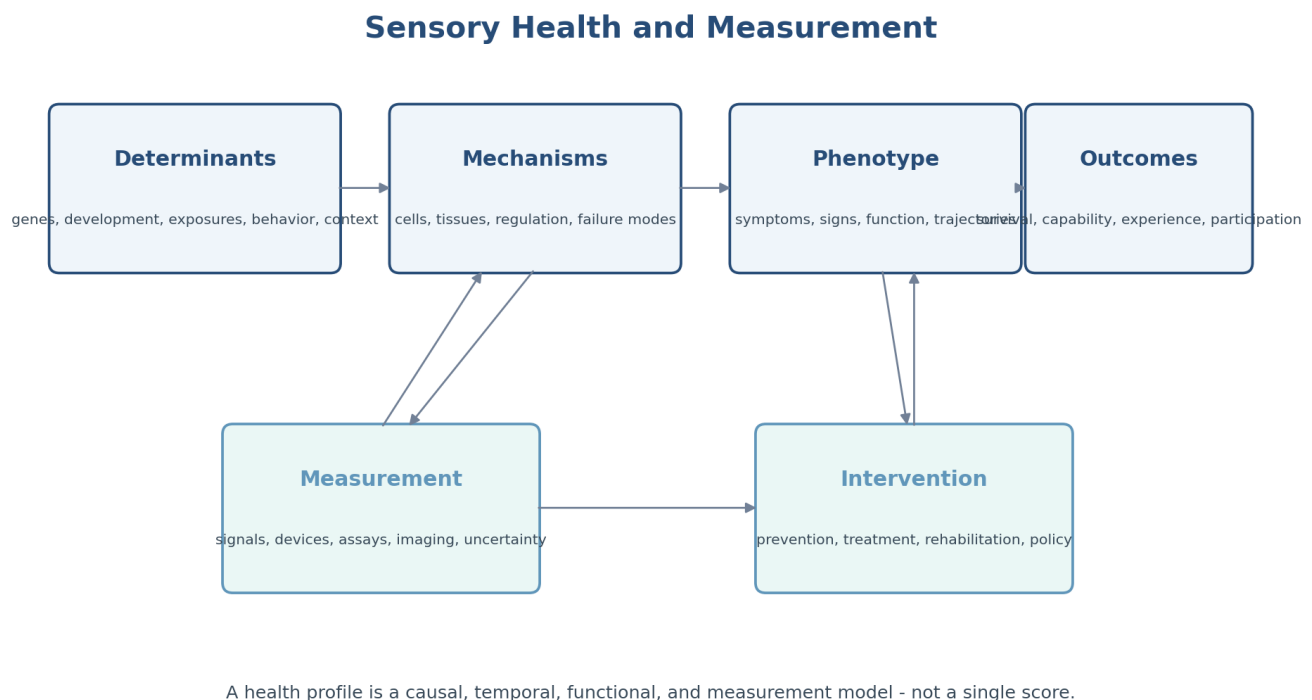
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# Executive synthesis

Sensory health is the capacity to detect, discriminate and interpret relevant information well enough to communicate, navigate, learn, work and participate. It cannot be reduced to a threshold: acuity, contrast, speech-in-noise, adaptation, multisensory integration, fatigue, environment and personal goals all matter. Device outputs require calibration to physical stimuli and validation against real-world function.

## Central organizing principle

Integrate peripheral sensory organs, neural coding, perception, development, function and assistive technology into a unified measurement and intervention framework.



**Figure 1.1:** The causal and measurement architecture used throughout this volume.

## 1.1 What a complete profile requires

- A multiscale account from molecules and cells through tissues, organs, whole-person function, and context.
- A longitudinal account of baseline, reserve under challenge, variability, compensation, injury, recovery, and irreversible change.
- Separation of causal determinants from correlates, biomarkers, manifestations, and consequences.
- Explicit representation of measurement error, missingness, artifact, reference methods, population drift, and clinical actionability.

- Integration with other atlas domains rather than treating one specialty as a closed system.

**Safety boundary**

Sudden vision or hearing loss, painful red eye with vision change, chemical eye injury, retinal-detachment symptoms, new focal neurologic signs, severe vertigo with neurologic deficits or temporal-arteritis pattern require urgent evaluation. Do not let a consumer screen delay care.

# Historical development and changing models

The history of a domain is not decorative. It explains why present-day categories, instruments, thresholds, specialties, and blind spots look the way they do. Technical advances often changed the object of medicine itself: what could be seen, counted, localized, or treated became the preferred unit of explanation.

| Period                          | Development                                                                                                               | What it made possible                                                                         | Residual limitation                                                                   |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Early traditions                | Optics, hearing, balance and flavor were studied through observation, craft and philosophy.                               | Created observational language for sensory health and practical care.                         | Mechanisms and categories mixed empirical findings with culture-specific theory.      |
| Anatomical and pathological era | Anatomy of eye, ear, nerves and brain localized sensory pathways.                                                         | Localized structure and tissue change.                                                        | Static anatomy poorly represented dynamics, perception, function and context.         |
| Instrumental and laboratory era | Lenses, ophthalmoscopy, audiometers, electrophysiology and imaging quantified organ and pathway function.                 | Turned previously subjective phenomena into reproducible signals and images.                  | Thresholds and reference populations sometimes became mistaken for health itself.     |
| Molecular and therapeutic era   | Phototransduction, cochlear mechanics, receptor genetics and implants enabled mechanism-specific care.                    | Connected cells, genes, microbes and signaling to targeted prevention and treatment.          | Mechanistic specificity often exceeded evidence of patient-important benefit.         |
| Current multimodal era          | OCT, digital sensing, gene therapy, neuroprosthetics and accessible technologies link microstructure to participation.    | Combines imaging, physiology, function, patient report, longitudinal sensing and computation. | Bias, interoperability, overdiagnosis, access and real-world utility remain limiting. |
| Systems interpretation          | Sensory health is now best modeled across tissue state, neural perception, behavior, environment, participation and care. | Supports primary-home and cross-volume reasoning.                                             | Training, billing and datasets still separate closely coupled phenomena.              |

## 2.0.1 Optics, hearing, balance and flavor were studied through observation, craft and philosophy.

**Scientific consequence.** Created observational language for sensory health and practical care.

**What remained unresolved.** Mechanisms and categories mixed empirical findings with culture-specific theory.

*Evidence anchors:* <sup>[3,5]</sup>

## 2.0.2 Anatomy of eye, ear, nerves and brain localized sensory pathways.

**Scientific consequence.** Localized structure and tissue change.

**What remained unresolved.** Static anatomy poorly represented dynamics, perception, function and context.

*Evidence anchors:* <sup>[3,5]</sup>

## 2.0.3 Lenses, ophthalmoscopy, audiometers, electrophysiology and imaging quantified organ and pathway function.

**Scientific consequence.** Turned previously subjective phenomena into reproducible signals and images.

**What remained unresolved.** Thresholds and reference populations sometimes became mistaken for health itself.

*Evidence anchors:* <sup>[3,5]</sup>

## 2.0.4 Phototransduction, cochlear mechanics, receptor genetics and implants enabled mechanism-specific care.

**Scientific consequence.** Connected cells, genes, microbes and signaling to targeted prevention and treatment.

**What remained unresolved.** Mechanistic specificity often exceeded evidence of patient-important benefit.

*Evidence anchors:* <sup>[3,5]</sup>

## 2.0.5 OCT, digital sensing, gene therapy, neuroprosthetics and accessible technologies link microstructure to participation.

**Scientific consequence.** Combines imaging, physiology, function, patient report, longitudinal sensing and computation.

**What remained unresolved.** Bias, interoperability, overdiagnosis, access and real-world utility remain limiting.

*Evidence anchors:* <sup>[3,5]</sup>

## 2.0.6 Sensory health is now best modeled across tissue state, neural perception, behavior, environment, participation and care.

**Scientific consequence.** Supports primary-home and cross-volume reasoning.

**What remained unresolved.** Training, billing and datasets still separate closely coupled phenomena.

*Evidence anchors:* <sup>[1]</sup>

# Normal foundations: structure, physiology, and regulation

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Normal physiology is represented here as regulated ranges and adaptive control, not as one fixed number. Many variables are intentionally dynamic, tissue-specific, state-dependent, and coupled to other systems. Disease can arise from primary damage, maladaptive compensation, loss of reserve, mistimed regulation, or mismatch between organism and environment.

## 3.0.1 Optical image formation and accommodation

**Core principle.** Cornea and lens focus light onto retina while pupils and accommodation regulate image quality and illumination.

**Control architecture.** Curvature, refractive index, axial length, ciliary control, tear film and neural feedback.

**Failure modes.** Refractive error, cataract, corneal irregularity, glare or focus loss.

**Observable consequences.** Acuity, refraction, wavefront, contrast, slit-lamp and task performance.

**Cross-system coupling.** Neurologic, developmental, metabolic and environmental systems.

*Evidence anchors:* [3,4,10]

## 3.0.2 Retinal phototransduction and visual coding

**Core principle.** Rods/cones convert photons to neural signals processed by retinal circuits before optic-nerve transmission.

**Control architecture.** Photopigments, adaptation, retinal metabolism/perfusion and synaptic networks.

**Failure modes.** Retinal degeneration, vascular injury, toxic/metabolic damage or detachment.

**Observable consequences.** Fundus/OCT, fields, color/contrast, ERG and function.

**Cross-system coupling.** Cardiovascular, metabolic, neurologic and circadian systems.

*Evidence anchors:* [2,3,13]

## 3.0.3 Auditory transduction

**Core principle.** External/middle ear transmit sound; cochlear hair cells and neural circuits encode frequency, intensity and timing.

**Control architecture.** Mechanical impedance, basilar membrane, active outer-hair-cell gain, synapses and central processing.

**Failure modes.** Conductive, sensorineural, neural or central hearing loss and tinnitus.

**Observable consequences.** Otoscopy, tympanometry, audiometry, OAE, ABR and communication function.

**Cross-system coupling.** Language, cognition, social participation and occupation.

*Evidence anchors:* [5,6,10]

### 3.0.4 Vestibular sensing and balance

**Core principle.** Semicircular canals and otoliths detect head motion and gravity, integrated with vision and proprioception.

**Control architecture.** Hair-cell transduction, vestibulo-ocular reflex, cerebellar/brainstem adaptation and multisensory weighting.

**Failure modes.** Vertigo, oscillopsia, imbalance, falls or motion intolerance.

**Observable consequences.** Eye movements, positional tests, vHIT/calorics, posturography and gait.

**Cross-system coupling.** Nervous, musculoskeletal, visual and autonomic systems.

*Evidence anchors:* [5,8,12]

### 3.0.5 Olfaction and taste

**Core principle.** Chemical receptors encode volatile and soluble molecules, integrated with trigeminal sensation and flavor perception.

**Control architecture.** Receptor binding, epithelial turnover, airflow/saliva, cranial nerves and central learning.

**Failure modes.** Anosmia, dysgeusia, distortion, nutrition/safety effects or neurodegenerative clues.

**Observable consequences.** Validated odor/taste tests, examination, imaging selectively and function.

**Cross-system coupling.** Respiratory, oral, neurologic, nutrition and infection systems.

*Evidence anchors:* [10,11,16]

### 3.0.6 Sensory adaptation and dynamic range

**Core principle.** Systems adjust gain to background intensity and context to preserve responsiveness.

**Control architecture.** Photochemical, mechanical, synaptic, cortical and attentional gain control.

**Failure modes.** Overload, reduced contrast, recruitment, hyperacusis, photophobia or sensory intolerance.

**Observable consequences.** Thresholds across backgrounds, adaptation curves, subjective tolerance and real-world tasks.

**Cross-system coupling.** Brain/behavior, migraine, autism/neurodevelopment and environment.

*Evidence anchors:* [11,20,22]

### 3.0.7 Development and critical periods

**Core principle.** Sensory input shapes neural circuits, language, motor control and learning during sensitive developmental windows.

**Control architecture.** Genetics, maturation, experience, deprivation and plasticity.

**Failure modes.** Amblyopia, language delay, persistent processing limits or compensatory adaptation.

**Observable consequences.** Age-appropriate screening, behavior, electrophysiology and developmental function.

**Cross-system coupling.** Life course, education, cognition and social systems.

*Evidence anchors:* [4,6,17]

### 3.0.8 Aging and reserve

**Core principle.** Lens, retina, cochlea, neural processing and multisensory integration change with age, interacting with cognition and environment.

**Control architecture.** Cell loss, oxidative/metabolic stress, vascular disease, cumulative exposure and reduced plasticity.

**Failure modes.** Vision/hearing loss, falls, isolation, cognitive load and loss of independence.

**Observable consequences.** Sensory tests, communication, mobility, cognition and participation trajectories.

**Cross-system coupling.** Aging, cardiovascular/metabolic, cognition and social connection.

*Evidence anchors:* [6,18,19]

### 3.0.9 Perception, attention and meaning

**Core principle.** Sensory organs provide signals that are interpreted by predictive neural systems shaped by attention, memory and context.

**Control architecture.** Bottom-up evidence, top-down prediction, salience, learning and cross-modal integration.

**Failure modes.** Hallucination/illusion, neglect, functional symptoms, tinnitus distress or sensory overload.

**Observable consequences.** Organ testing plus cognitive/neurologic/psychological assessment and functional impact.

**Cross-system coupling.** Brain/cognition, mental health and psychosocial domains.

*Evidence anchors:* <sup>[11,20,21]</sup>

### 3.1 Useful quantitative relationships

These equations are compact models. Their variables require operational definitions, units, boundary conditions, and a statement of when the approximation fails.

| Relationship                 | Expression                                    | Interpretation                                                                    | Limits                                                                                    |
|------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Decibel sound pressure level | $L_p = 20 \log_{10}(p/p_0)$                   | Logarithmic level relative to a reference pressure.                               | Weighting, time averaging, spectrum and exposure duration determine biological relevance. |
| Thin-lens relationship       | $1/f = 1/d_o + 1/d_i$                         | Relates focal length to object and image distance in an idealized optical system. | The living eye has multiple refracting surfaces, aberrations and accommodation.           |
| Vestibulo-ocular reflex gain | $Gain = \frac{eye\ velocity}{head\ velocity}$ | Ideal compensatory eye motion approximately opposes head motion.                  | Frequency, direction, calibration and corrective saccades must be considered.             |

# Determinants, heterogeneity, and causal architecture

No universal percentage can divide health into genetics and environment. Contributions vary by outcome, age, population, exposure distribution, measurement scale, and causal estimand. Genetic liability can alter sensitivity to an exposure; exposures can alter gene regulation; both operate through development, behavior, social conditions, and access to prevention or care.

| Determinant                             | Main pathways                                                                                                                                                                                                                                       | Leverage                                                                                                                                               | Measurement                                                                                                                                                                          | Interpretive caution                                                                                                                                                    |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Genetic architecture and family history | Monogenic variants, structural variation, mitochondrial inheritance, polygenic liability, and pleiotropy can alter baseline structure, regulatory set points, reserve, treatment response, and vulnerability within sensory health and measurement. | Genotype is not modifiable, but surveillance, exposure reduction, reproductive counseling, and targeted therapy may change outcomes.                   | Three-generation pedigree when relevant; phenotype-first testing; chromosomal, panel, exome/genome, repeat-expansion, mitochondrial, or pharmacogenomic assays selected by question. | Heritability is population- and environment-specific; ancestry is not a biological proxy for race; variants of uncertain significance must not be treated as diagnoses. |
| Development and life course             | Organ formation, critical periods, pubertal transitions, pregnancy, cumulative damage, repair capacity, and aging alter both baseline function and response to later exposures.                                                                     | Preconception health, maternal care, early nutrition, vaccination, education, injury prevention, and age-appropriate screening can shift trajectories. | Gestational and birth history, developmental milestones, longitudinal growth/function, reproductive history, age-related reserve and frailty.                                        | Adult snapshots can misattribute developmental effects to current behavior; cohort and survivor effects distort age comparisons.                                        |
| Nutrition and metabolism                | Energy balance, protein and micronutrient sufficiency, dietary pattern, glycemic and lipid exposure, microbiome substrates, and toxic contaminants affect tissue maintenance and signaling.                                                         | Food environment, dietary pattern, supplementation for verified deficiency, treatment of metabolic disease, and reduction of harmful exposures.        | Dietary history with known reporting error, anthropometry, relevant biomarkers, body composition, and repeated metabolic measures.                                                   | Single nutrients rarely capture the causal pattern; reverse causation and socioeconomic confounding are common.                                                         |

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| <b>Determinant</b>                                      | <b>Main pathways</b>                                                                                                                                                            | <b>Leverage</b>                                                                                                                                                  | <b>Measurement</b>                                                                                                                                           | <b>Interpretive caution</b>                                                                                                                   |
|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Physical activity, conditioning, and sedentary behavior | Mechanical loading, perfusion, mitochondrial adaptation, autonomic regulation, insulin sensitivity, mood, sleep, and functional reserve.                                        | Progressive activity and exercise adapted to disability, symptoms, safety, and goals; reduction of prolonged sedentary time.                                     | Device-measured movement plus context, intensity, posture, strength, cardiorespiratory or task-specific capacity.                                            | Wearables misclassify some activities and disability patterns; association with health partly reflects baseline capacity.                     |
| Sleep, circadian timing, and recovery                   | Hormonal timing, autonomic balance, immune regulation, memory, tissue repair, appetite, and exposure opportunity are synchronized by sleep and circadian systems.               | Regular timing, treatment of sleep disorders, light management, work-schedule redesign, and reduction of disruptive substances when appropriate.                 | History, sleep diary, actigraphy, validated questionnaires, physiologic sleep testing when indicated.                                                        | Consumer sleep stages are model outputs, not direct polysomnographic equivalents; illness can cause poor sleep and vice versa.                |
| Stress, emotion, and behavior                           | Threat appraisal, sympathetic and neuroendocrine activation, inflammation, coping behavior, adherence, substance use, and social interaction can alter exposure and physiology. | Safety, psychological treatment, social support, work redesign, coping skills, treatment of mental disorders, and reduction of chronic uncontrollable stressors. | Contextual history, validated reported measures, repeated physiology when useful, and assessment of material stressors rather than one generic stress score. | Biomarkers are nonspecific; blaming individuals for structural stress is scientifically and ethically wrong.                                  |
| Social, economic, cultural, and care conditions         | Income, education, housing, work, discrimination, language, legal status, relationships, culture, care access, and treatment burden shape exposures and options.                | Policy, income and housing support, accessible care, trust, communication, transportation, accommodations, and community resources.                              | Person-reported context, geospatial and administrative data, access/quality measures, and participatory qualitative methods.                                 | Do not convert social disadvantage into an immutable patient trait; structural causes require structural interventions.                       |
| Environmental and occupational exposures                | Air, water, heat, noise, radiation, chemicals, pathogens, ergonomic load, shift work, and injury hazards act through dose, timing, route, mixtures, and vulnerability.          | Source control, substitution, ventilation, personal protection, regulation, surveillance, and climate adaptation.                                                | Job and location history, direct and modeled exposure, biomarkers where valid, and uncertainty in mixtures and cumulative dose.                              | Personal sensors may not estimate biologically effective dose; exposure misclassification usually attenuates but can also bias unpredictably. |

Continued on next page

| Determinant                                     | Main pathways                                                                                                                                                     | Leverage                                                                                                                         | Measurement                                                                                                                 | Interpretive caution                                                                                                       |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Health care, medication, and iatrogenic effects | Access, diagnostic delay, treatment quality, polypharmacy, procedures, device complications, antimicrobial exposure, and discontinuity can prevent or cause harm. | Medication reconciliation, deprescribing when appropriate, safer systems, shared decisions, guideline adaptation, and follow-up. | Longitudinal medication/procedure record, adverse events, care transitions, adherence, treatment burden, and patient goals. | Confounding by indication makes treated people appear sicker; absence of care data can be mistaken for absence of disease. |

#### 4.0.1 Genetic architecture and family history

**Mechanistic pathways.** Monogenic variants, structural variation, mitochondrial inheritance, polygenic liability, and pleiotropy can alter baseline structure, regulatory set points, reserve, treatment response, and vulnerability within sensory health and measurement.

**Potentially modifiable components.** Genotype is not modifiable, but surveillance, exposure reduction, reproductive counseling, and targeted therapy may change outcomes.

**How to measure it.** Three-generation pedigree when relevant; phenotype-first testing; chromosomal, panel, exome/genome, repeat-expansion, mitochondrial, or pharmacogenomic assays selected by question.

**Causal and equity cautions.** Heritability is population- and environment-specific; ancestry is not a biological proxy for race; variants of uncertain significance must not be treated as diagnoses.

*Evidence anchors:* <sup>[30]</sup>

#### 4.0.2 Development and life course

**Mechanistic pathways.** Organ formation, critical periods, pubertal transitions, pregnancy, cumulative damage, repair capacity, and aging alter both baseline function and response to later exposures.

**Potentially modifiable components.** Preconception health, maternal care, early nutrition, vaccination, education, injury prevention, and age-appropriate screening can shift trajectories.

**How to measure it.** Gestational and birth history, developmental milestones, longitudinal growth/function, reproductive history, age-related reserve and frailty.

**Causal and equity cautions.** Adult snapshots can misattribute developmental effects to current behavior; cohort and survivor effects distort age comparisons.

*Evidence anchors:* <sup>[17]</sup>

#### 4.0.3 Nutrition and metabolism

**Mechanistic pathways.** Energy balance, protein and micronutrient sufficiency, dietary pattern, glycemic and lipid exposure, microbiome substrates, and toxic contaminants affect tissue maintenance and signaling.

**Potentially modifiable components.** Food environment, dietary pattern, supplementation for verified deficiency, treatment of metabolic disease, and reduction of harmful exposures.

**How to measure it.** Dietary history with known reporting error, anthropometry, relevant biomarkers, body composition, and repeated metabolic measures.

**Causal and equity cautions.** Single nutrients rarely capture the causal pattern; reverse causation and socioeconomic confounding are common.

#### 4.0.4 Physical activity, conditioning, and sedentary behavior

**Mechanistic pathways.** Mechanical loading, perfusion, mitochondrial adaptation, autonomic regulation, insulin sensitivity, mood, sleep, and functional reserve.

**Potentially modifiable components.** Progressive activity and exercise adapted to disability, symptoms, safety, and goals; reduction of prolonged sedentary time.

**How to measure it.** Device-measured movement plus context, intensity, posture, strength, cardiorespiratory or task-specific capacity.

**Causal and equity cautions.** Wearables misclassify some activities and disability patterns; association with health partly reflects baseline capacity.

#### 4.0.5 Sleep, circadian timing, and recovery

**Mechanistic pathways.** Hormonal timing, autonomic balance, immune regulation, memory, tissue repair, appetite, and exposure opportunity are synchronized by sleep and circadian systems.

**Potentially modifiable components.** Regular timing, treatment of sleep disorders, light management, work-schedule redesign, and reduction of disruptive substances when appropriate.

**How to measure it.** History, sleep diary, actigraphy, validated questionnaires, physiologic sleep testing when indicated.

**Causal and equity cautions.** Consumer sleep stages are model outputs, not direct polysomnographic equivalents; illness can cause poor sleep and vice versa.

#### 4.0.6 Stress, emotion, and behavior

**Mechanistic pathways.** Threat appraisal, sympathetic and neuroendocrine activation, inflammation, coping behavior, adherence, substance use, and social interaction can alter exposure and physiology.

**Potentially modifiable components.** Safety, psychological treatment, social support, work redesign, coping skills, treatment of mental disorders, and reduction of chronic uncontrollable stressors.

**How to measure it.** Contextual history, validated reported measures, repeated physiology when useful, and assessment of material stressors rather than one generic stress score.

**Causal and equity cautions.** Biomarkers are nonspecific; blaming individuals for structural stress is scientifically and ethically wrong.

#### 4.0.7 Social, economic, cultural, and care conditions

**Mechanistic pathways.** Income, education, housing, work, discrimination, language, legal status, relationships, culture, care access, and treatment burden shape exposures and options.

**Potentially modifiable components.** Policy, income and housing support, accessible care, trust, communication, transportation, accommodations, and community resources.

**How to measure it.** Person-reported context, geospatial and administrative data, access/quality measures, and participatory qualitative methods.

**Causal and equity cautions.** Do not convert social disadvantage into an immutable patient trait; structural causes require structural interventions.

#### 4.0.8 Environmental and occupational exposures

**Mechanistic pathways.** Air, water, heat, noise, radiation, chemicals, pathogens, ergonomic load, shift work, and injury hazards act through dose, timing, route, mixtures, and vulnerability.

**Potentially modifiable components.** Source control, substitution, ventilation, personal protection, regulation, surveillance, and climate adaptation.

**How to measure it.** Job and location history, direct and modeled exposure, biomarkers where valid, and uncertainty in mixtures and cumulative dose.

**Causal and equity cautions.** Personal sensors may not estimate biologically effective dose; exposure misclassification usually attenuates but can also bias unpredictably.

*Evidence anchors:* <sup>[29]</sup>

#### 4.0.9 Health care, medication, and iatrogenic effects

**Mechanistic pathways.** Access, diagnostic delay, treatment quality, polypharmacy, procedures, device complications, antimicrobial exposure, and discontinuity can prevent or cause harm.

**Potentially modifiable components.** Medication reconciliation, deprescribing when appropriate, safer systems, shared decisions, guideline adaptation, and follow-up.

**How to measure it.** Longitudinal medication/procedure record, adverse events, care transitions, adherence, treatment burden, and patient goals.

**Causal and equity cautions.** Confounding by indication makes treated people appear sicker; absence of care data can be mistaken for absence of disease.

*Evidence anchors:* <sup>[31]</sup>

# Major disorders, failure modes, and clinically important phenotypes

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The profiles below are not a replacement for a diagnostic manual. They are organized to connect mechanism, phenotype, measurement, intervention, and uncertainty. The primary unit is the sensory impairment, organ disorder, perceptual phenotype, or participation limitation, but cross-system processes and functional effects are retained.

## 5.0.1 Refractive error and amblyopia

**Definition and phenotype.** Defocus from eye geometry/optics; amblyopia is reduced visual development not fully corrected by optics alone.

**Pathophysiology.** Myopia/hyperopia/astigmatism alter retinal image; early unequal input changes cortical visual development.

**Determinants and modifiers.** Genetics, near/outdoor exposure, development, access to screening and correction.

**Assessment.** Age-appropriate acuity, cycloplegic refraction when indicated, binocular alignment and ocular health.

**Intervention logic.** Glasses/contact/lens or refractive options; early occlusion/optical strategies for amblyopia.

**Known limits and open questions.** Myopia-control comparative durability and population implementation continue to evolve.

**Suite links.** 07 development; 11 screening; 15 access.

*Evidence anchors:* <sup>[3,4,17]</sup>

## 5.0.2 Cataract and lens opacity

**Definition and phenotype.** Loss of lens transparency causing blur, glare and reduced contrast.

**Pathophysiology.** Protein aggregation, oxidative/metabolic injury and structural change scatter/absorb light.

**Determinants and modifiers.** Age, diabetes, UV, smoking, steroids, trauma and access to surgery.

**Assessment.** Visual function, slit-lamp, refraction and exclusion of posterior disease.

**Intervention logic.** Surgery with lens implantation when functional benefit outweighs risk; optimize systemic/exposure factors.

**Known limits and open questions.** Timing is preference/function-based rather than one acuity threshold.

**Suite links.** 03D diabetes; 09 UV; 12 surgical access.

*Evidence anchors:* <sup>[3,4]</sup>

### 5.0.3 Glaucoma

**Definition and phenotype.** Progressive optic neuropathy with characteristic structural and field loss, often but not always associated with intraocular pressure.

**Pathophysiology.** Pressure-related and pressure-independent susceptibility damages retinal ganglion axons.

**Determinants and modifiers.** Age, family/ancestry, pressure, corneal properties, vascular factors and access.

**Assessment.** Tonometry, optic nerve/OCT, visual fields, angle and serial progression.

**Intervention logic.** Pressure lowering by drops, laser or surgery; adherence and lifelong monitoring.

**Known limits and open questions.** Screening and progression prediction vary; device measurements are not interchangeable.

**Suite links.** 03A vascular; 13 measurement.

*Evidence anchors:* <sup>[3,4]</sup>

### 5.0.4 Retinal vascular and diabetic eye disease

**Definition and phenotype.** Microvascular leakage, ischemia, neovascularization or occlusion injures retina.

**Pathophysiology.** Hyperglycemia, hypertension, thrombosis and endothelial dysfunction disrupt retinal perfusion/barrier.

**Determinants and modifiers.** Diabetes duration/control, blood pressure, kidney disease, smoking and care access.

**Assessment.** Dilated imaging, photography/OCT/OCT-A and angiography selectively; systemic assessment.

**Intervention logic.** Risk-factor management, intravitreal therapy, laser or surgery by phenotype.

**Known limits and open questions.** Imaging markers and visual function can diverge; screening access remains unequal.

**Suite links.** 03D diabetes; 03A vascular; 03F kidney.

*Evidence anchors:* [2,3,13]

### 5.0.5 Age-related macular degeneration and inherited retinal degeneration

**Definition and phenotype.** Central or diffuse retinal degeneration causing progressive visual loss.

**Pathophysiology.** Complement/lipid/oxidative pathways, aging, neovascularization or genetic photoreceptor/RPE defects.

**Determinants and modifiers.** Age, genetics, smoking, diet patterns and specific inherited variants.

**Assessment.** Fundus/OCT, fields/function, angiography and genetic counseling/testing where actionable.

**Intervention logic.** Smoking cessation, selected supplements by phenotype, anti-VEGF for neovascular disease, low-vision care and gene therapy for narrow inherited indications.

**Known limits and open questions.** Biomarker prediction and broad gene/retinal restoration remain frontier.

**Suite links.** 07 aging; 14 gene therapy; 10 assistive technology.

*Evidence anchors:* [3,7,30]

### 5.0.6 Corneal and ocular-surface disease

**Definition and phenotype.** Tear, epithelial, inflammatory, infectious or structural disease causing discomfort, optical degradation and injury.

**Pathophysiology.** Tear instability, barrier disruption, immune/microbial injury and corneal shape/scarring.

**Determinants and modifiers.** Contact lenses, environment, autoimmune disease, surgery, infection and medications.

**Assessment.** Symptoms plus slit lamp, staining, tear tests, topography and microbiology when severe/atypical.

**Intervention logic.** Lubrication/environment, anti-inflammatory or antimicrobial therapy, protective procedures and transplantation selectively.

**Known limits and open questions.** Dry-eye tests correlate imperfectly with symptoms.

**Suite links.** 03H immune; 03J barrier; 09 environment.

*Evidence anchors:* [3,14]

### 5.0.7 Hearing loss

**Definition and phenotype.** Reduced audibility or speech understanding from conductive, cochlear, neural or central mechanisms.

**Pathophysiology.** Obstruction/middle-ear mechanics, hair-cell/synapse loss, nerve or processing deficits.

**Determinants and modifiers.** Age, noise, genetics, infection, ototoxic drugs, prematurity and care access.

**Assessment.** Otoscopy, audiometry, speech testing, tympanometry/OAE/ABR and communication goals.

**Intervention logic.** Treat reversible causes; hearing aids, implants, communication/rehabilitation and environmental accommodations.

**Known limits and open questions.** Pure-tone thresholds incompletely represent real-world speech and fatigue.

**Suite links.** 09 noise; 10 assistive technology; 06 social connection.

*Evidence anchors:* [5,6,7]

### 5.0.8 Tinnitus and sound intolerance

**Definition and phenotype.** Perception of sound without an external source or reduced tolerance to ordinary sound.

**Pathophysiology.** Peripheral loss and central gain/prediction networks interact with attention, arousal and distress.

**Determinants and modifiers.** Hearing loss, noise, drugs, jaw/neck factors, stress and sleep.

**Assessment.** Red flags/laterality/pulsatility, hearing/ear exam, targeted imaging and distress/function.

**Intervention logic.** Education, hearing treatment, sound and cognitive-behavioral approaches; treat specific vascular/structural causes.

**Known limits and open questions.** No single biomarker or universal cure; distress is not proportional to loudness.

**Suite links.** 05 brain/behavior; 03A pulsatile causes; 03I jaw/neck.

*Evidence anchors:* [5,20,22]

### 5.0.9 Otitis and middle-ear disease

**Definition and phenotype.** Inflammation, fluid, infection or structural dysfunction of external/middle ear causing pain and conductive loss.

**Pathophysiology.** Ventilation dysfunction, microbes, barrier injury and fluid alter sound transmission.

**Determinants and modifiers.** Age, respiratory infection, water/trauma, smoking exposure, craniofacial factors and access.

**Assessment.** Otoscopy, tympanometry/audiometry and microbiology selectively.

**Intervention logic.** Pain care, observation or antimicrobials by syndrome, drainage/tubes or surgery for selected persistent/complicated disease.

**Known limits and open questions.** Overdiagnosis and antibiotic overuse occur when visualization is poor.

**Suite links.** 08 infection; 03B upper airway; 03L craniofacial.

*Evidence anchors:* [5,15]

### 5.0.10 Vestibular disorders

**Definition and phenotype.** Peripheral or central disturbances causing vertigo, imbalance, oscillopsia or motion sensitivity.

**Pathophysiology.** Canaliths, vestibular hypofunction, migraine, inflammation, vascular or central lesions alter multisensory estimates.

**Determinants and modifiers.** Age, infection, migraine, ototoxicity, head injury and neurologic disease.

**Assessment.** Timing/triggers, neurologic exam, positional/eye-movement tests and targeted vestibular/imaging studies.

**Intervention logic.** Canalith repositioning, vestibular rehabilitation, cause-specific treatment and fall prevention; limit chronic suppressants.

**Known limits and open questions.** Symptoms overlap and test sensitivity depends on timing and compensation.

**Suite links.** 03C nervous; 03I balance; 10 rehab.

*Evidence anchors:* [5,8,11]

### 5.0.11 Smell and taste disorders

**Definition and phenotype.** Loss, reduction, distortion or hallucination of chemical senses affecting nutrition, safety and quality of life.

**Pathophysiology.** Nasal obstruction/epithelial injury, cranial nerve/central disease, medications or oral/salivary factors.

**Determinants and modifiers.** Infection, trauma, sinonasal disease, aging, smoking, neurodegeneration and drugs.

**Assessment.** Validated psychophysical testing, ENT/oral/neuro exam and imaging/labs selectively.

**Intervention logic.** Treat cause, smell training for selected post-infectious loss, safety/nutrition strategies and symptom support.

**Known limits and open questions.** Mechanistic biomarkers and effective restoration options remain limited.

**Suite links.** 03B airway; 03L oral; 03C nervous; 08 infection.

*Evidence anchors:* [8,11,16]

### 5.0.12 Sensory processing and perceptual disorders

**Definition and phenotype.** Difficulty detecting, integrating or assigning meaning to sensory input despite variable peripheral-organ function.

**Pathophysiology.** Neural network development/injury, attention and predictive processing alter perception and behavior.

**Determinants and modifiers.** Neurodevelopment, brain injury, migraine, psychiatric/neurologic disease and environment.

**Assessment.** Rule out peripheral deficits; neurologic, cognitive, occupational and functional assessment.

**Intervention logic.** Environmental adaptation, rehabilitation and treatment of defined neurologic/mental-health conditions.

**Known limits and open questions.** Broad “processing disorder” labels may lack standardized boundaries.

**Suite links.** 05 cognition/behavior; 07 development; 10 rehabilitation.

*Evidence anchors:* [8,20,21]

### 5.0.13 Deafblindness and multisensory impairment

**Definition and phenotype.** Combined sensory loss with effects greater than the sum of individual impairments.

**Pathophysiology.** Reduced redundant input alters communication, orientation, learning and access.

**Determinants and modifiers.** Congenital syndromes, aging, infection, trauma and inequitable services.

**Assessment.** Integrated functional communication, mobility and sensory evaluation with person/caregiver goals.

**Intervention logic.** Early intervention, communication systems, assistive technology, environmental and social support.

**Known limits and open questions.** Standard single-sense tests underestimate participation burden.

**Suite links.** 10 disability/AT; 15 rights; 06 social health.

*Evidence anchors:* [4,6,9]

### 5.0.14 Ocular and auditory emergencies

**Definition and phenotype.** Acute vision/hearing loss, chemical injury, retinal detachment, angle closure, temporal arteritis, sudden sensorineural loss or neurologic syndromes require urgent pathways.

**Pathophysiology.** Ischemia, pressure, detachment, inflammation, trauma or infection can cause irreversible damage.

**Determinants and modifiers.** Age, vascular/immune risk, trauma/exposure and delayed recognition.

**Assessment.** Time of onset, focused organ and neurologic exam, pressure/imaging/labs by syndrome.

**Intervention logic.** Immediate mechanism-specific specialist/emergency treatment.

**Known limits and open questions.** Consumer screening can falsely reassure and time-to-care dominates outcome.

**Suite links.** 03A vascular; 03H immune; 12 emergency systems.

*Evidence anchors:* [3,5,31]

# Assessment, clinical reasoning, and longitudinal profiling

Assessment is a staged inference process. The right test depends on the pretest question, urgency, expected change in management, burden, and the consequences of false reassurance or false alarm. A complete profile combines history, examination, validated measurements, function, trajectories, and context.

| Stage                                           | Purpose                                                                                                            | Components                                                                                                                                     | Output                                                                                       | Common failure                                                                                                     |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| 1. Question and urgency                         | Define the decision, symptom, screening target, research hypothesis, and time horizon; identify emergencies first. | Chief concern and goals; red flags; prior probability; relevant context and current sensory health and measurement risks.                      | Urgent action, focused differential, or a prespecified measurement plan.                     | Testing without a question creates incidental findings and false precision.                                        |
| 2. History and lived phenotype                  | Characterize onset, pattern, triggers, severity, variability, function, treatment response, and personal meaning.  | Symptoms, timeline, development, family history, exposures, medications, behavior, sleep, psychosocial and occupational context.               | A longitudinal problem representation with hypotheses and missing information.               | Administrative codes and checklists can erase chronology, context, and patient priorities.                         |
| 3. Examination and basic measurements           | Test hypotheses using standardized observation and low-burden validated measures.                                  | Relevant physical, neurologic, behavioral, functional, and vital-sign examination; quality-controlled basic tests.                             | Localization, severity estimate, safety screen, and indication for further testing.          | Nonstandard technique and unblinded interpretation create systematic error.                                        |
| 4. Targeted laboratory, imaging, and physiology | Confirm or refute a mechanism, stage severity, identify treatable causes, or establish a reference baseline.       | Select tests by expected information gain; document protocol, reference range, preanalytic conditions, and contrast/radiation/invasive burden. | Mechanism-informed diagnosis, prognostic information, or a revised probability distribution. | Reference intervals are not universal decision thresholds; incidental abnormalities may not explain the phenotype. |
| 5. Function and challenge testing               | Measure reserve, adaptation, and real-world impact not visible at rest.                                            | Exercise, task, stress, provocation, simulated activity, ecological monitoring, and patient-reported function as appropriate.                  | Capacity, symptom coupling, recovery, and safety limits.                                     | Challenge protocols may not match daily demands and can be unsafe without screening.                               |

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| Stage                       | Purpose                                                                                                           | Components                                                                                                                     | Output                                                                   | Common failure                                                                        |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 6. Longitudinal integration | Distinguish stable trait, reversible state, episodic event, progression, treatment effect, and measurement drift. | Repeated measures with synchronized context, interventions, adherence, quality, and prespecified meaningful-change thresholds. | Trajectory, response, relapse, and need for escalation or de-escalation. | Regression to the mean and changing devices can masquerade as improvement or decline. |

### 6.0.1 1. Question and urgency

**Purpose.** Define the decision, symptom, screening target, research hypothesis, and time horizon; identify emergencies first.

**Core components.** Chief concern and goals; red flags; prior probability; relevant context and current sensory health and measurement risks.

**Interpretation.** Urgent action, focused differential, or a prespecified measurement plan.

**Failure modes.** Testing without a question creates incidental findings and false precision.

### 6.0.2 2. History and lived phenotype

**Purpose.** Characterize onset, pattern, triggers, severity, variability, function, treatment response, and personal meaning.

**Core components.** Symptoms, timeline, development, family history, exposures, medications, behavior, sleep, psychosocial and occupational context.

**Interpretation.** A longitudinal problem representation with hypotheses and missing information.

**Failure modes.** Administrative codes and checklists can erase chronology, context, and patient priorities.

*Evidence anchors:* <sup>[22]</sup>

### 6.0.3 3. Examination and basic measurements

**Purpose.** Test hypotheses using standardized observation and low-burden validated measures.

**Core components.** Relevant physical, neurologic, behavioral, functional, and vital-sign examination; quality-controlled basic tests.

**Interpretation.** Localization, severity estimate, safety screen, and indication for further testing.

**Failure modes.** Nonstandard technique and unblinded interpretation create systematic error.

#### 6.0.4 4. Targeted laboratory, imaging, and physiology

**Purpose.** Confirm or refute a mechanism, stage severity, identify treatable causes, or establish a reference baseline.

**Core components.** Select tests by expected information gain; document protocol, reference range, preanalytic conditions, and contrast/radiation/invasive burden.

**Interpretation.** Mechanism-informed diagnosis, prognostic information, or a revised probability distribution.

**Failure modes.** Reference intervals are not universal decision thresholds; incidental abnormalities may not explain the phenotype.

#### 6.0.5 5. Function and challenge testing

**Purpose.** Measure reserve, adaptation, and real-world impact not visible at rest.

**Core components.** Exercise, task, stress, provocation, simulated activity, ecological monitoring, and patient-reported function as appropriate.

**Interpretation.** Capacity, symptom coupling, recovery, and safety limits.

**Failure modes.** Challenge protocols may not match daily demands and can be unsafe without screening.

#### 6.0.6 6. Longitudinal integration

**Purpose.** Distinguish stable trait, reversible state, episodic event, progression, treatment effect, and measurement drift.

**Core components.** Repeated measures with synchronized context, interventions, adherence, quality, and prespecified meaningful-change thresholds.

**Interpretation.** Trajectory, response, relapse, and need for escalation or de-escalation.

**Failure modes.** Regression to the mean and changing devices can masquerade as improvement or decline.

# Measurement and device landscape

A device does not measure health directly. It senses a physical or chemical proxy, transforms it, samples it, rejects or retains artifact, and maps the result to a claim. The claim may concern detection, diagnosis, severity, risk, response, or function; each requires a different validation design.

## Measurement is an inference chain



Validation must test every link, including population, setting, motion, missingness, drift, and downstream use.

**Figure 7.1:** Generic inference chain from latent biology to a clinical or research claim.

| Modality                                         | Invasiveness        | Signal / sensor                                                                                    | Outputs                                                           | Strengths                                  | Limitations                                                                   |
|--------------------------------------------------|---------------------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|--------------------------------------------|-------------------------------------------------------------------------------|
| Visual acuity, refraction and contrast testing   | Noninvasive         | Psychophysical identification/detection under controlled optics, distance, luminance and contrast. | High/low-contrast resolution and refractive correction.           | Direct functional output and scalable.     | Learning, language, luminance, chart, crowding and effort effects.            |
| Slit-lamp, ophthalmoscopy and fundus photography | Noninvasive         | Magnified illumination and reflected-light imaging of anterior/posterior eye.                      | Structural lesions, media, optic nerve and retina.                | Rich direct anatomy.                       | Media opacity, field, operator and image quality.                             |
| Tonometry and corneal biomechanics               | Noninvasive/Contact | Applanation, rebound, air-puff or other force-deformation estimates.                               | Intraocular pressure and biomechanical proxies.                   | Essential glaucoma risk/treatment measure. | Corneal thickness/rigidity, technique, time and device bias.                  |
| Optical coherence tomography and angiography     | Noninvasive         | Low-coherence interferometry and motion contrast.                                                  | Retinal/optic-nerve layers, fluid and microvascular flow proxies. | Micron-scale repeatable structure.         | Segmentation, motion, signal strength, device non-equivalence and false flow. |

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| <b>Modality</b>                                          | <b>Invasiveness</b>     | <b>Signal / sensor</b>                                                              | <b>Outputs</b>                                                           | <b>Strengths</b>                                | <b>Limitations</b>                                                                    |
|----------------------------------------------------------|-------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------|-------------------------------------------------|---------------------------------------------------------------------------------------|
| Visual fields, electroretinography and evoked potentials | Noninvasive             | Psychophysical spatial detection and retinal/cortical electrical responses.         | Field sensitivity, retinal function and pathway conduction.              | Functional localization beyond anatomy.         | Fatigue/learning, electrode/noise, fixation and limited specificity.                  |
| Audiometry and speech-in-noise testing                   | Noninvasive             | Calibrated acoustic stimuli and behavioral responses.                               | Air/bone thresholds, speech recognition and real-world listening proxy.  | Core functional hearing map.                    | Booth/transducer calibration, attention/language and limited ecological validity.     |
| Tympanometry and acoustic reflexes                       | Noninvasive             | Ear-canal pressure/sound measurements of middle-ear impedance and reflex response.  | Canal volume, pressure, compliance and reflex patterns.                  | Rapid conductive-pathway assessment.            | Seal, anatomy, infant frequency and interpretation limits.                            |
| Otoacoustic emissions and auditory brainstem response    | Noninvasive             | Cochlear-generated ear-canal sound and scalp electrophysiology to acoustic stimuli. | Outer-hair-cell function and auditory pathway timing/threshold estimate. | Objective newborn and nonbehavioral assessment. | Noise, middle-ear state, electrode artifact and does not equal perception.            |
| Video-oculography, vHIT and caloric/rotational testing   | Noninvasive             | Infrared eye tracking during head motion, temperature or rotation stimuli.          | Vestibulo-ocular gain, saccades and canal-specific asymmetry.            | Objective dynamic vestibular function.          | Goggle slip, calibration, compensation and frequency-specific disagreement.           |
| Posturography and gait/balance sensing                   | Noninvasive             | Force platform and motion/inertial measures during stance/tasks.                    | Sway, sensory weighting, limits and fall-risk proxies.                   | Integrated functional balance.                  | Strategy, fear, learning and limited diagnostic specificity.                          |
| Psychophysical smell and taste testing                   | Noninvasive             | Standardized odor/tastant identification, threshold and discrimination.             | Chemical-sense function and pattern.                                     | More valid than self-report alone.              | Culture/language, nasal/oral state, adaptation and stimulus standardization.          |
| Imaging: CT, MRI and targeted angiography                | Noninvasive to contrast | Anatomic and functional imaging of orbit, temporal bone, pathways and brain.        | Structural, inflammatory, vascular, neoplastic and central causes.       | Essential for selected red flags and planning.  | Incidental findings, radiation/contrast, cost and low yield without pretest question. |

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| Modality                                                      | Invasiveness                 | Signal / sensor                                                                                 | Outputs                                             | Strengths                                              | Limitations                                                                             |
|---------------------------------------------------------------|------------------------------|-------------------------------------------------------------------------------------------------|-----------------------------------------------------|--------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Hearing aids, cochlear implants and device telemetry          | Noninvasive/implantable      | Microphones, DSP, electrical stimulation and logged use/impedance.                              | Amplification/stimulation, fitting state and usage. | Restores access to sound/communication for many users. | Fit, cognition, noise, battery/connectivity, surgery and unequal access.                |
| Low-vision, orientation and multisensory assistive technology | Noninvasive                  | Optical/electronic magnification, screen access, haptics, navigation and communication systems. | Task support and participation.                     | Directly addresses lived function.                     | Training, cost, interoperability, abandonment and inaccessible environments.            |
| Consumer sensory screening and AI                             | Noninvasive soft-ware/device | Phone displays/cameras, earbuds/microphones and algorithmic classification.                     | Screening score or triage.                          | Potential reach and repeated measurement.              | Uncalibrated hardware, ambient noise/light, access bias, false reassurance and privacy. |

### 7.0.1 Visual acuity, refraction and contrast testing

**Measured quantity and transducer.** Psychophysical identification/detection under controlled optics, distance, luminance and contrast.

**Typical outputs.** High/low-contrast resolution and refractive correction.

**Strengths.** Direct functional output and scalable.

**Limitations and artifacts.** Learning, language, luminance, chart, crowding and effort effects.

**Reference and validation strategy.** Standard charts/illumination, repeatability and task validity.

**Clinical/research maturity.** Established.

*Evidence anchors:* <sup>[3,4]</sup>

### 7.0.2 Slit-lamp, ophthalmoscopy and fundus photography

**Measured quantity and transducer.** Magnified illumination and reflected-light imaging of anterior/posterior eye.

**Typical outputs.** Structural lesions, media, optic nerve and retina.

**Strengths.** Rich direct anatomy.

**Limitations and artifacts.** Media opacity, field, operator and image quality.

**Reference and validation strategy.** Standard acquisition, expert reference and longitudinal grading.

**Clinical/research maturity.** Established.

*Evidence anchors:* <sup>[3]</sup>

### 7.0.3 Tonometry and corneal biomechanics

**Measured quantity and transducer.** Applanation, rebound, air-puff or other force-deformation estimates.

**Typical outputs.** Intraocular pressure and biomechanical proxies.

**Strengths.** Essential glaucoma risk/treatment measure.

**Limitations and artifacts.** Corneal thickness/rigidity, technique, time and device bias.

**Reference and validation strategy.** Calibration, method comparison and progression/outcome linkage.

**Clinical/research maturity.** Established with method limits.

*Evidence anchors:* <sup>[3,25]</sup>

### 7.0.4 Optical coherence tomography and angiography

**Measured quantity and transducer.** Low-coherence interferometry and motion contrast.

**Typical outputs.** Retinal/optic-nerve layers, fluid and microvascular flow proxies.

**Strengths.** Micron-scale repeatable structure.

**Limitations and artifacts.** Segmentation, motion, signal strength, device non-equivalence and false flow.

**Reference and validation strategy.** Phantoms, manual adjudication, cross-device and clinical progression validation.

**Clinical/research maturity.** Established/advanced.

*Evidence anchors:* <sup>[3,23]</sup>

### 7.0.5 Visual fields, electroretinography and evoked potentials

**Measured quantity and transducer.** Psychophysical spatial detection and retinal/cortical electrical responses.

**Typical outputs.** Field sensitivity, retinal function and pathway conduction.

**Strengths.** Functional localization beyond anatomy.

**Limitations and artifacts.** Fatigue/learning, electrode/noise, fixation and limited specificity.

**Reference and validation strategy.** Standard stimuli/electrodes, repeatability and disease reference.

**Clinical/research maturity.** Established specialist.

*Evidence anchors:* <sup>[3,27]</sup>

## 7.0.6 Audiometry and speech-in-noise testing

**Measured quantity and transducer.** Calibrated acoustic stimuli and behavioral responses.

**Typical outputs.** Air/bone thresholds, speech recognition and real-world listening proxy.

**Strengths.** Core functional hearing map.

**Limitations and artifacts.** Booth/transducer calibration, attention/language and limited ecological validity.

**Reference and validation strategy.** Acoustic standards, test-retest and communication outcomes.

**Clinical/research maturity.** Established.

*Evidence anchors:* <sup>[5,6]</sup>

## 7.0.7 Tympanometry and acoustic reflexes

**Measured quantity and transducer.** Ear-canal pressure/sound measurements of middle-ear impedance and reflex response.

**Typical outputs.** Canal volume, pressure, compliance and reflex patterns.

**Strengths.** Rapid conductive-pathway assessment.

**Limitations and artifacts.** Seal, anatomy, infant frequency and interpretation limits.

**Reference and validation strategy.** Calibration cavities, protocol and audiologic correlation.

**Clinical/research maturity.** Established.

*Evidence anchors:* <sup>[5]</sup>

### 7.0.8 Otoacoustic emissions and auditory brainstem response

**Measured quantity and transducer.** Cochlear-generated ear-canal sound and scalp electrophysiology to acoustic stimuli.

**Typical outputs.** Outer-hair-cell function and auditory pathway timing/threshold estimate.

**Strengths.** Objective newborn and nonbehavioral assessment.

**Limitations and artifacts.** Noise, middle-ear state, electrode artifact and does not equal perception.

**Reference and validation strategy.** Calibrated stimuli, artifact rules and behavioral follow-up.

**Clinical/research maturity.** Established.

*Evidence anchors:* [5,27]

### 7.0.9 Video-oculography, vHIT and caloric/rotational testing

**Measured quantity and transducer.** Infrared eye tracking during head motion, temperature or rotation stimuli.

**Typical outputs.** Vestibulo-ocular gain, saccades and canal-specific asymmetry.

**Strengths.** Objective dynamic vestibular function.

**Limitations and artifacts.** Goggle slip, calibration, compensation and frequency-specific disagreement.

**Reference and validation strategy.** Motion rigs, expert protocol and functional outcome correlation.

**Clinical/research maturity.** Established specialist.

*Evidence anchors:* [5,23]

### 7.0.10 Posturography and gait/balance sensing

**Measured quantity and transducer.** Force platform and motion/inertial measures during stance/tasks.

**Typical outputs.** Sway, sensory weighting, limits and fall-risk proxies.

**Strengths.** Integrated functional balance.

**Limitations and artifacts.** Strategy, fear, learning and limited diagnostic specificity.

**Reference and validation strategy.** Calibration, repeatability and prospective fall/rehab validation.

**Clinical/research maturity.** Established/research.

*Evidence anchors:* [8,12]

### 7.0.11 Psychophysical smell and taste testing

**Measured quantity and transducer.** Standardized odor/tastant identification, threshold and discrimination.

**Typical outputs.** Chemical-sense function and pattern.

**Strengths.** More valid than self-report alone.

**Limitations and artifacts.** Culture/language, nasal/oral state, adaptation and stimulus standardization.

**Reference and validation strategy.** Validated local norms and longitudinal repeatability.

**Clinical/research maturity.** Established specialist/research.

*Evidence anchors:* [11,16]

### 7.0.12 Imaging: CT, MRI and targeted angiography

**Measured quantity and transducer.** Anatomic and functional imaging of orbit, temporal bone, pathways and brain.

**Typical outputs.** Structural, inflammatory, vascular, neoplastic and central causes.

**Strengths.** Essential for selected red flags and planning.

**Limitations and artifacts.** Incidental findings, radiation/contrast, cost and low yield without pretest question.

**Reference and validation strategy.** Indication-specific protocol and specialist reference.

**Clinical/research maturity.** Established.

*Evidence anchors:* [3,11]

### 7.0.13 Hearing aids, cochlear implants and device telemetry

**Measured quantity and transducer.** Microphones, DSP, electrical stimulation and logged use/impedance.

**Typical outputs.** Amplification/stimulation, fitting state and usage.

**Strengths.** Restores access to sound/communication for many users.

**Limitations and artifacts.** Fit, cognition, noise, battery/connectivity, surgery and unequal access.

**Reference and validation strategy.** Electroacoustic bench tests, speech/function and long-term use/outcomes.

**Clinical/research maturity.** Established.

*Evidence anchors:* [5,7,27]

### 7.0.14 Low-vision, orientation and multisensory assistive technology

**Measured quantity and transducer.** Optical/electronic magnification, screen access, haptics, navigation and communication systems.

**Typical outputs.** Task support and participation.

**Strengths.** Directly addresses lived function.

**Limitations and artifacts.** Training, cost, interoperability, abandonment and inaccessible environments.

**Reference and validation strategy.** User-centered task trials and real-world participation outcomes.

**Clinical/research maturity.** Established to emerging.

*Evidence anchors:* [7,9]

### 7.0.15 Consumer sensory screening and AI

**Measured quantity and transducer.** Phone displays/cameras, earbuds/microphones and algorithmic classification.

**Typical outputs.** Screening score or triage.

**Strengths.** Potential reach and repeated measurement.

**Limitations and artifacts.** Uncalibrated hardware, ambient noise/light, access bias, false reassurance and privacy.

**Reference and validation strategy.** Hardware eligibility checks, external comparison, subgroup and clinical utility.

**Clinical/research maturity.** Emerging; screening only within validated use.

*Evidence anchors:* [23,24,28]

# Engineering architectures, signal quality, and validation

Buildable systems should begin with an explicit intended use, measurand, reference method, population, environment, and failure policy. Prototype plausibility is not clinical validity. Raw data, quality channels, timing, calibration state, firmware version, and provenance should be retained so that errors can be reconstructed rather than hidden by a polished score.

| Subsystem / design issue                   | Requirements                                                                                                                                                         | Dominant risks                                                                                                                    | Verification                                                                                                                |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Intended use and measurand                 | Define the precise sensory health and measurement quantity, population, setting, user, decision, sampling interval, and acceptable uncertainty.                      | A broad wellness claim can hide several incompatible clinical claims.                                                             | Written claims matrix; trace each claim to design inputs, reference method, dataset, and acceptance criterion.              |
| Sensor and transduction                    | Characterize sensitivity, range, noise, bandwidth, cross-sensitivity, hysteresis, saturation, drift, temperature dependence, and biocompatible interface.            | A plausible-looking waveform can arise from motion, pressure, ambient energy, crosstalk, contamination, or algorithmic smoothing. | Calibrated physical/chemical standards, simulators, controlled perturbations, and failure injection across operating range. |
| Timing and synchronization                 | Use explicit clocks, timestamp provenance, drift characterization, event markers, and synchronization uncertainty for multimodal data.                               | Millisecond-to-minute drift can create false lead-lag physiology or destroy event alignment.                                      | Long-duration clock tests, known pulse trains, dropout/restart tests, and end-to-end latency measurement.                   |
| Analog, acquisition, and digital integrity | Document gain, filtering, ADC performance, anti-aliasing, resolution, dynamic range, saturation recovery, packet loss, and compression.                              | Hidden filters distort morphology and phase; quantization and clipping may be mislabeled as biology.                              | Frequency response, noise spectral density, step response, injected signals, checksum and data-loss testing.                |
| Calibration and traceability               | Specify calibration hierarchy, reference uncertainty, intervals, field checks, lot/device effects, and recalibration triggers.                                       | Model calibration to one device lot, site, or operator may not transport.                                                         | Traceable standards, multi-device studies, interoperator studies, environmental chambers, and blinded comparison.           |
| Artifact and quality channels              | Measure contact, motion, pressure, temperature, ambient conditions, specimen quality, and operator state rather than inferring quality only from the primary signal. | Artifact may correlate with disease, disability, skin tone, age, or setting and create biased missingness.                        | Artifact challenge set, subgroup performance, quality-controlled exclusion rules, and sensitivity analyses.                 |

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| Subsystem / design issue                      | Requirements                                                                                                                                               | Dominant risks                                                                                               | Verification                                                                                                                             |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Algorithms and uncertainty                    | Separate signal processing, feature extraction, model inference, confidence, calibration, abstention, and human review.                                    | Data leakage, spectrum bias, shortcut learning, nonstationarity, overfitting, and automation bias.           | Locked external validation, temporal/geographic transport tests, calibration curves, decision analysis, adversarial and failure testing. |
| Safety, usability, cybersecurity, and privacy | Risk management, electrical and mechanical safety, biocompatibility, human factors, secure update, least privilege, audit, consent, and data minimization. | Use error, unintended data disclosure, unsafe remote control, vulnerable update paths, and treatment delays. | Hazard analysis, formative/summative usability, penetration and abuse-case testing, incident response drills, and postmarket monitoring. |

### 8.0.1 Intended use and measurand

**Design requirements.** Define the precise sensory health and measurement quantity, population, setting, user, decision, sampling interval, and acceptable uncertainty.

**Failure modes.** A broad wellness claim can hide several incompatible clinical claims.

**Bench verification.** Written claims matrix; trace each claim to design inputs, reference method, dataset, and acceptance criterion.

**Human validation boundary.** Do not enroll people until the measurement purpose and risk controls are reviewable.

*Evidence anchors:* [23,26]

### 8.0.2 Sensor and transduction

**Design requirements.** Characterize sensitivity, range, noise, bandwidth, cross-sensitivity, hysteresis, saturation, drift, temperature dependence, and biocompatible interface.

**Failure modes.** A plausible-looking waveform can arise from motion, pressure, ambient energy, crosstalk, contamination, or algorithmic smoothing.

**Bench verification.** Calibrated physical/chemical standards, simulators, controlled perturbations, and failure injection across operating range.

**Human validation boundary.** Bench testing should precede contact with participants.

*Evidence anchors:* [26,27]

### 8.0.3 Timing and synchronization

**Design requirements.** Use explicit clocks, timestamp provenance, drift characterization, event markers, and synchronization uncertainty for multimodal data.

**Failure modes.** Millisecond-to-minute drift can create false lead-lag physiology or destroy event alignment.

**Bench verification.** Long-duration clock tests, known pulse trains, dropout/restart tests, and end-to-end latency measurement.

**Human validation boundary.** Store both device time and host/reference time.

### 8.0.4 Analog, acquisition, and digital integrity

**Design requirements.** Document gain, filtering, ADC performance, anti-aliasing, resolution, dynamic range, saturation recovery, packet loss, and compression.

**Failure modes.** Hidden filters distort morphology and phase; quantization and clipping may be mislabeled as biology.

**Bench verification.** Frequency response, noise spectral density, step response, injected signals, checksum and data-loss testing.

**Human validation boundary.** Preserve raw or minimally processed channels where ethically and technically possible.

*Evidence anchors:* <sup>[25]</sup>

### 8.0.5 Calibration and traceability

**Design requirements.** Specify calibration hierarchy, reference uncertainty, intervals, field checks, lot/device effects, and recalibration triggers.

**Failure modes.** Model calibration to one device lot, site, or operator may not transport.

**Bench verification.** Traceable standards, multi-device studies, interoperator studies, environmental chambers, and blinded comparison.

**Human validation boundary.** Calibration data and version must travel with each observation.

*Evidence anchors:* <sup>[25]</sup>

### 8.0.6 Artifact and quality channels

**Design requirements.** Measure contact, motion, pressure, temperature, ambient conditions, specimen quality, and operator state rather than inferring quality only from the primary signal.

**Failure modes.** Artifact may correlate with disease, disability, skin tone, age, or setting and create biased missingness.

**Bench verification.** Artifact challenge set, subgroup performance, quality-controlled exclusion rules, and sensitivity analyses.

**Human validation boundary.** Quality rejection must not silently exclude populations.

*Evidence anchors:* <sup>[28]</sup>

### 8.0.7 Algorithms and uncertainty

**Design requirements.** Separate signal processing, feature extraction, model inference, confidence, calibration, abstention, and human review.

**Failure modes.** Data leakage, spectrum bias, shortcut learning, nonstationarity, overfitting, and automation bias.

**Bench verification.** Locked external validation, temporal/geographic transport tests, calibration curves, decision analysis, adversarial and failure testing.

**Human validation boundary.** A model should abstain when inputs are outside validated conditions.

*Evidence anchors:* <sup>[24,28]</sup>

### 8.0.8 Safety, usability, cybersecurity, and privacy

**Design requirements.** Risk management, electrical and mechanical safety, biocompatibility, human factors, secure update, least privilege, audit, consent, and data minimization.

**Failure modes.** Use error, unintended data disclosure, unsafe remote control, vulnerable update paths, and treatment delays.

**Bench verification.** Hazard analysis, formative/summative usability, penetration and abuse-case testing, incident response drills, and postmarket monitoring.

**Human validation boundary.** Research exemptions do not remove the ethical duty to minimize foreseeable harm.

*Evidence anchors:* <sup>[27]</sup>

#### Human-contact engineering rule

Do not connect a human participant to a mains-referenced oscilloscope, programmer, charger, or non-medical ground path. Bench characterization should use simulators and isolated fixtures. Human research requires documented electrical safety, risk management, consent, privacy controls, and appropriate oversight.

# Prevention, treatment, rehabilitation, and system intervention

Interventions are classified by target and level: upstream determinants, exposure reduction, behavioral support, prevention, diagnosis-linked treatment, acute rescue, chronic management, rehabilitation, assistive technology, and palliative or supportive care. Net benefit depends on baseline risk, absolute effect, burden, interactions, access, adherence, and the outcomes valued by the person.

| Intervention                                      | Level                     | Target / mechanism                                                                                                            | Evidence profile                                                                                            | Risks / burdens                                                      | Monitoring                                                            |
|---------------------------------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------------|
| Early detection and correction                    | Prevention/rehabilitation | Identify and treatable sensory loss and provide optical, medical, surgical or assistive correction during meaningful windows. | Strong for refractive error, cataract, many hearing losses and childhood critical periods.                  | Screening cascades, false reassurance and unequal access.            | Functional outcomes, development, adherence and downstream care.      |
| Exposure and vascular/metabolic risk reduction    | Prevention                | Control noise/ototoxins/UV/trauma and systemic vascular/metabolic drivers.                                                    | Prevents selected hearing, eye and balance injury; systemic management reduces retinal disease progression. | PPE burden and exposure inequity; treatment adverse effects.         | Exposure dose, hearing/retinal trajectory and systemic targets.       |
| Pharmacologic and local biologic therapy          | Treatment                 | Treat infection/inflammation, lower pressure or inhibit neovascular signaling.                                                | Strong for defined ocular/ear diseases.                                                                     | Local/systemic toxicity, resistance, adherence and procedure burden. | Organ function/structure, infection, pressure and adverse effects.    |
| Surgery and implants                              | Definitive/assistive      | Restore optical media/drainage/anatomy or electrically stimulate auditory pathways.                                           | Large functional benefit in selected cataract, glaucoma, retinal, middle-ear and hearing-loss states.       | Infection, failure, revision, device dependence and access.          | Complications, device function, sensory and participation outcomes.   |
| Vision, hearing and vestibular rehabilitation     | Rehabilitation            | Train compensation, communication, gaze/balance, navigation and task adaptation.                                              | Improves function and participation despite persistent organ loss.                                          | Training burden and service gaps.                                    | Task ability, falls, communication, fatigue, participation and goals. |
| Assistive and accessible-environment intervention | Participation             | Provide aids, captions, sign/tactile communication, contrast, lighting, acoustic and navigation access.                       | Often determines real-world disability more than impairment magnitude alone.                                | Cost, maintenance, stigma and inaccessible systems.                  | Use, task success, independence, education/work/social participation. |

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| Intervention                                                 | Level                    | Target / mechanism                                                                            | Evidence profile                                                                                     | Risks / burdens                                                              | Monitoring                                                                             |
|--------------------------------------------------------------|--------------------------|-----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Canalith repositioning and cause-specific vestibular therapy | Treatment/rehabilitation | Mobilize displaced otoconia or retrain gaze/postural compensation.                            | High benefit for correctly diagnosed positional vertigo; rehabilitation supports hypofunction.       | Transient symptoms, falls and misdiagnosis of central disease.               | Provocation resolution, gaze/balance and recurrence.                                   |
| Tinnitus and sensory-distress management                     | Whole-person             | Treat hearing/medical contributors and reduce attention, arousal and functional interference. | Hearing devices and cognitive-behavioral approaches help selected outcomes; no universal sound cure. | Commercial overclaim and invalidation.                                       | Distress, sleep, communication, function and goals.                                    |
| Communication and family/education support                   | Developmental            | Support language, literacy, communication mode, caregiver skill and school/work access.       | Critical to developmental and participation outcomes.                                                | Delayed services, cultural conflict and one-size approaches.                 | Language/learning, relationships, school/work and self-determined communication goals. |
| Gene, retinal and neural restoration technologies            | Frontier                 | Replace genes/cells, protect circuits or create visual/auditory percepts.                     | Established only for narrow indications; broad restoration remains experimental.                     | Immune/surgical harm, durability, unrealistic expectations, cost and equity. | Objective function, patient-important use, adverse events and long-term durability.    |

### 9.0.1 Early detection and correction

**Target and mechanism.** Identify treatable sensory loss and provide optical, medical, surgical or assistive correction during meaningful windows.

**Evidence and expected outcome.** Strong for refractive error, cataract, many hearing losses and childhood critical periods.

**Risks, burden, and equity.** Screening cascades, false reassurance and unequal access.

**Response monitoring.** Functional outcomes, development, adherence and downstream care.

*Evidence anchors:* [4,6,17]

### 9.0.2 Exposure and vascular/metabolic risk reduction

**Target and mechanism.** Control noise/ototoxins/UV/trauma and systemic vascular/metabolic drivers.

**Evidence and expected outcome.** Prevents selected hearing, eye and balance injury; systemic management reduces retinal disease progression.

**Risks, burden, and equity.** PPE burden and exposure inequity; treatment adverse effects.

**Response monitoring.** Exposure dose, hearing/retinal trajectory and systemic targets.

*Evidence anchors:* [2,13,29]

### 9.0.3 Pharmacologic and local biologic therapy

**Target and mechanism.** Treat infection/inflammation, lower pressure or inhibit neovascular signaling.

**Evidence and expected outcome.** Strong for defined ocular/ear diseases.

**Risks, burden, and equity.** Local/systemic toxicity, resistance, adherence and procedure burden.

**Response monitoring.** Organ function/structure, infection, pressure and adverse effects.

*Evidence anchors:* [3,15]

### 9.0.4 Surgery and implants

**Target and mechanism.** Restore optical media/drainage/anatomy or electrically stimulate auditory pathways.

**Evidence and expected outcome.** Large functional benefit in selected cataract, glaucoma, retinal, middle-ear and hearing-loss states.

**Risks, burden, and equity.** Infection, failure, revision, device dependence and access.

**Response monitoring.** Complications, device function, sensory and participation outcomes.

*Evidence anchors:* [3,5,7]

### 9.0.5 Vision, hearing and vestibular rehabilitation

**Target and mechanism.** Train compensation, communication, gaze/balance, navigation and task adaptation.

**Evidence and expected outcome.** Improves function and participation despite persistent organ loss.

**Risks, burden, and equity.** Training burden and service gaps.

**Response monitoring.** Task ability, falls, communication, fatigue, participation and goals.

*Evidence anchors:* [8]

### 9.0.6 Assistive and accessible-environment intervention

**Target and mechanism.** Provide aids, captions, sign/tactile communication, contrast, lighting, acoustic and navigation access.

**Evidence and expected outcome.** Often determines real-world disability more than impairment magnitude alone.

**Risks, burden, and equity.** Cost, maintenance, stigma and inaccessible systems.

**Response monitoring.** Use, task success, independence, education/work/social participation.

*Evidence anchors:* [7,9]

### 9.0.7 Canalith repositioning and cause-specific vestibular therapy

**Target and mechanism.** Move displaced otoconia or retrain gaze/postural compensation.

**Evidence and expected outcome.** High benefit for correctly diagnosed positional vertigo; rehabilitation supports hypofunction.

**Risks, burden, and equity.** Transient symptoms, falls and misdiagnosis of central disease.

**Response monitoring.** Provocation resolution, gaze/balance and recurrence.

*Evidence anchors:* [5,8]

### 9.0.8 Tinnitus and sensory-distress management

**Target and mechanism.** Treat hearing/medical contributors and reduce attention, arousal and functional interference.

**Evidence and expected outcome.** Hearing devices and cognitive-behavioral approaches help selected outcomes; no universal sound cure.

**Risks, burden, and equity.** Commercial overclaim and invalidation.

**Response monitoring.** Distress, sleep, communication, function and goals.

*Evidence anchors:* [5,22]

### 9.0.9 Communication and family/education support

**Target and mechanism.** Support language, literacy, communication mode, caregiver skill and school/work access.

**Evidence and expected outcome.** Critical to developmental and participation outcomes.

**Risks, burden, and equity.** Delayed services, cultural conflict and one-size approaches.

**Response monitoring.** Language/learning, relationships, school/work and self-determined communication goals.

*Evidence anchors:* [4,6,9]

### 9.0.10 Gene, retinal and neural restoration technologies

**Target and mechanism.** Replace genes/cells, protect circuits or create visual/auditory percepts.

**Evidence and expected outcome.** Established only for narrow indications; broad restoration remains experimental.

**Risks, burden, and equity.** Immune/surgical harm, durability, unrealistic expectations, cost and equity.

**Response monitoring.** Objective function, patient-important use, adverse events and long-term durability.

*Evidence anchors:* <sup>[30]</sup>

# What is known, not known, and frequently overstated

Uncertainty is not a single residual. It can arise from biological variability, measurement error, selection, confounding, limited follow-up, inadequate reference standards, endpoint substitution, poor external validity, causal ambiguity, or value disagreement. The correct response is to identify the uncertainty and design the next measurement or study.

| Question                                                                       | What is established                                                                                                   | What remains uncertain                                                                                                                           | Why unresolved                                                                                                             | Decisive next step                                                                                                                      |
|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Can hidden hearing or visual dysfunction be measured before thresholds change? | Synaptic, contrast and processing deficits can occur despite conventional thresholds.                                 | Which tests are reliable, specific and treatment-actionable.                                                                                     | No single gold standard and symptoms are multidimensional.                                                                 | Longitudinal exposure/outcome cohorts and intervention-response validation.                                                             |
| Do sensory-restoration neural interfaces improve daily life?                   | Implants can create useful percepts in selected pathways.                                                             | Long-term naturalistic benefit, cognitive load and equitable usability for broader indications.                                                  | Laboratory discrimination tasks overstate ecological function.                                                             | Pragmatic trials with participation, burden, device reliability and user-defined outcomes.                                              |
| How much risk is genetic versus environmental?                                 | Both inherited variation and environments contribute, often through interaction and mediation across the life course. | No single percentage applies to all outcomes, populations, ages, or definitions within sensory health and measurement.                           | Heritability changes with exposure distribution; family studies mix genes and shared environment; causal pathways overlap. | Large diverse longitudinal cohorts with family structure, repeated exposure measurement, molecular data, and explicit causal estimands. |
| Can one composite score summarize health?                                      | Composite scores can rank a defined outcome in a defined population and may improve workflow.                         | Whether a score preserves causal meaning, transportability, subgroup calibration, and patient-important outcomes outside development conditions. | Different variables compensate, interact, and have different treatment implications; weights drift over time.              | Externally validated transparent components, decision-curve analysis, recalibration, and reporting of disagreement among domains.       |

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| Question                              | What is established                                                                                                          | What remains uncertain                                                                                                  | Why unresolved                                                                                    | Decisive next step                                                                                                   |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| When is earlier detection beneficial? | Early detection helps when there is a detectable preclinical phase, an effective earlier intervention, and acceptable harms. | Net benefit for many incidental, molecular, wearable, and multi-cancer signals.                                         | Lead-time, length, overdiagnosis, cascade testing, anxiety, and treatment harms can dominate.     | Randomized or strong quasi-experimental studies with patient-important outcomes and explicit overdiagnosis analysis. |
| Do digital biomarkers generalize?     | Some digital measures reliably quantify narrow behaviors or signals under controlled conditions.                             | Long-term meaning across devices, firmware, disability, skin characteristics, cultures, settings, and disease severity. | Proprietary processing, dataset shift, artifact-dependent missingness, and weak reference labels. | Open protocols, raw-data retention, standards, multisite external validation, and equity-centered failure analysis.  |
| What counts as meaningful change?     | Statistical change, measurement error, minimal detectable change, and patient-important change are distinct.                 | Thresholds for many biomarkers, sensors, and multidomain profiles.                                                      | Baseline severity, goals, adaptation, and treatment burden alter value.                           | Anchor-based longitudinal studies integrating function, symptoms, goals, adverse effects, and resource use.          |

### 10.0.1 Can hidden hearing or visual dysfunction be measured before thresholds change?

**Established knowledge.** Synaptic, contrast and processing deficits can occur despite conventional thresholds.

**Unresolved issue.** Which tests are reliable, specific and treatment-actionable.

**Why current evidence is insufficient.** No single gold standard and symptoms are multidimensional.

**Research that would change confidence.** Longitudinal exposure/outcome cohorts and intervention-response validation.

*Evidence anchors:* <sup>[3,5]</sup>

### 10.0.2 Do sensory-restoration neural interfaces improve daily life?

**Established knowledge.** Implants can create useful percepts in selected pathways.

**Unresolved issue.** Long-term naturalistic benefit, cognitive load and equitable usability for broader indications.

**Why current evidence is insufficient.** Laboratory discrimination tasks overstate ecological function.

**Research that would change confidence.** Pragmatic trials with participation, burden, device reliability and user-defined outcomes.

*Evidence anchors:* <sup>[7]</sup>

### 10.0.3 How much risk is genetic versus environmental?

**Established knowledge.** Both inherited variation and environments contribute, often through interaction and mediation across the life course.

**Unresolved issue.** No single percentage applies to all outcomes, populations, ages, or definitions within sensory health and measurement.

**Why current evidence is insufficient.** Heritability changes with exposure distribution; family studies mix genes and shared environment; causal pathways overlap.

**Research that would change confidence.** Large diverse longitudinal cohorts with family structure, repeated exposure measurement, molecular data, and explicit causal estimands.

*Evidence anchors:* <sup>[30]</sup>

### 10.0.4 Can one composite score summarize health?

**Established knowledge.** Composite scores can rank a defined outcome in a defined population and may improve workflow.

**Unresolved issue.** Whether a score preserves causal meaning, transportability, subgroup calibration, and patient-important outcomes outside development conditions.

**Why current evidence is insufficient.** Different variables compensate, interact, and have different treatment implications; weights drift over time.

**Research that would change confidence.** Externally validated transparent components, decision-curve analysis, recalibration, and reporting of disagreement among domains.

*Evidence anchors:* <sup>[22,28]</sup>

### 10.0.5 When is earlier detection beneficial?

**Established knowledge.** Early detection helps when there is a detectable preclinical phase, an effective earlier intervention, and acceptable harms.

**Unresolved issue.** Net benefit for many incidental, molecular, wearable, and multi-cancer signals.

**Why current evidence is insufficient.** Lead-time, length, overdiagnosis, cascade testing, anxiety, and treatment harms can dominate.

**Research that would change confidence.** Randomized or strong quasi-experimental studies with patient-important outcomes and explicit overdiagnosis analysis.

### 10.0.6 Do digital biomarkers generalize?

**Established knowledge.** Some digital measures reliably quantify narrow behaviors or signals under controlled conditions.

**Unresolved issue.** Long-term meaning across devices, firmware, disability, skin characteristics, cultures, settings, and disease severity.

**Why current evidence is insufficient.** Proprietary processing, dataset shift, artifact-dependent missingness, and weak reference labels.

**Research that would change confidence.** Open protocols, raw-data retention, standards, multisite external validation, and equity-centered failure analysis.

*Evidence anchors:* <sup>[23]</sup>

### 10.0.7 What counts as meaningful change?

**Established knowledge.** Statistical change, measurement error, minimal detectable change, and patient-important change are distinct.

**Unresolved issue.** Thresholds for many biomarkers, sensors, and multidomain profiles.

**Why current evidence is insufficient.** Baseline severity, goals, adaptation, and treatment burden alter value.

**Research that would change confidence.** Anchor-based longitudinal studies integrating function, symptoms, goals, adverse effects, and resource use.

*Evidence anchors:* <sup>[22]</sup>

# Research and development roadmap

The roadmap prioritizes work that changes decisions or reduces uncertainty, not merely work that generates more variables. Each project should declare the causal or measurement target, comparator, reference, population, sampling frame, missing-data plan, subgroup analysis, and stopping criteria before data collection.

| Priority                      | Question                                                                                                                                                                    | Minimum credible design                                                                                                                                             | Success criterion                                                                            | Stop / failure criterion                                                            |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 0. Multisensory reserve atlas | How do vision, hearing, vestibular and cognitive load interact under real tasks?                                                                                            | Diverse life-course cohort with calibrated stimuli, mobility/communication tasks, wearables and environmental contexts.                                             | Predicts falls, communication fatigue and rehabilitation response beyond single tests.       | Complexity without reproducible or actionable gain.                                 |
| 0. Consumer calibration layer | Can phones and earbuds self-verify hardware/environment before screening?                                                                                                   | Traceable light/acoustic/motion reference experiments and multicenter clinical comparison.                                                                          | Reliable abstention and bounded error across eligible hardware.                              | Unknown hardware creates silent bias or false reassurance.                          |
| 1. Reference map              | What are the normal ranges, state transitions, and reserve responses across age, sex, ancestry, disability, environment, and life stage for sensory health and measurement? | Prospective multi-site reference cohort; standardized protocols; repeated measures; raw data and quality channels; deliberate inclusion of underrepresented groups. | Reproducible distributions and state-dependent reference models with quantified uncertainty. | Convenience sample presented as universal normality.                                |
| 2. Causal determinants        | Which modifiable exposures cause clinically and functionally important outcomes, and in whom?                                                                               | Triangulation across randomized interventions, natural experiments, longitudinal cohorts, negative controls, genetics where appropriate, and mechanistic studies.   | Convergent effect estimates linked to plausible mechanisms and actionable interventions.     | Association-only result marketed as prevention guidance.                            |
| 3. Measurement validity       | Does the new device or biomarker measure the intended quantity under real use?                                                                                              | Bench metrology followed by blinded method comparison, repeatability, subgroup, artifact, transport, and longitudinal responsiveness studies.                       | Prespecified error and failure limits met across intended-use conditions.                    | High average correlation with systematic individual error or unreported exclusions. |
| 4. Clinical utility           | Does using the measurement improve decisions or outcomes compared with current care?                                                                                        | Randomized or pragmatic comparative study measuring downstream actions, benefits, harms, burden, equity, and cost.                                                  | Net benefit on patient-important outcomes with acceptable implementation burden.             | Improved prediction without changed decisions or outcomes.                          |

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| Priority                         | Question                                                                                     | Minimum credible design                                                                                                                                   | Success criterion                                                               | Stop / failure criterion                                                    |
|----------------------------------|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| 5. Longitudinal multimorbidity   | How does the domain interact with other conditions, treatments, and functional trajectories? | Person-centered longitudinal cohort or trial with common data model, medication and intervention timelines, function, participation, and competing risks. | Models that support coordinated care rather than isolated disease optimization. | Single-disease endpoint worsens overall burden or conflicts with goals.     |
| 6. Implementation and governance | Can effective interventions be delivered safely, equitably, affordably, and sustainably?     | Hybrid effectiveness-implementation studies, participatory design, workflow and workforce analysis, privacy/security review, and policy evaluation.       | Sustained adoption with monitored outcomes and reduced inequity.                | Pilot success depends on exceptional staffing or hidden externalized costs. |

### 11.0.1 0. Multisensory reserve atlas

**Question.** How do vision, hearing, vestibular and cognitive load interact under real tasks?

**Design.** Diverse life-course cohort with calibrated stimuli, mobility/communication tasks, wearables and environmental contexts.

**Success criterion.** Predicts falls, communication fatigue and rehabilitation response beyond single tests.

**Failure or stop criterion.** Complexity without reproducible or actionable gain.

### 11.0.2 0. Consumer calibration layer

**Question.** Can phones and earbuds self-verify hardware/environment before screening?

**Design.** Traceable light/acoustic/motion reference experiments and multicenter clinical comparison.

**Success criterion.** Reliable abstention and bounded error across eligible hardware.

**Failure or stop criterion.** Unknown hardware creates silent bias or false reassurance.

*Evidence anchors:* [23,28]

### 11.0.3 1. Reference map

**Question.** What are the normal ranges, state transitions, and reserve responses across age, sex, ancestry, disability, environment, and life stage for sensory health and measurement?

**Design.** Prospective multi-site reference cohort; standardized protocols; repeated measures; raw data and quality channels; deliberate inclusion of underrepresented groups.

**Success criterion.** Reproducible distributions and state-dependent reference models with quantified uncertainty.

**Failure or stop criterion.** Convenience sample presented as universal normality.

### 11.0.4 2. Causal determinants

**Question.** Which modifiable exposures cause clinically and functionally important outcomes, and in whom?

**Design.** Triangulation across randomized interventions, natural experiments, longitudinal cohorts, negative controls, genetics where appropriate, and mechanistic studies.

**Success criterion.** Convergent effect estimates linked to plausible mechanisms and actionable interventions.

**Failure or stop criterion.** Association-only result marketed as prevention guidance.

### 11.0.5 3. Measurement validity

**Question.** Does the new device or biomarker measure the intended quantity under real use?

**Design.** Bench metrology followed by blinded method comparison, repeatability, subgroup, artifact, transport, and longitudinal responsiveness studies.

**Success criterion.** Prespecified error and failure limits met across intended-use conditions.

**Failure or stop criterion.** High average correlation with systematic individual error or unreported exclusions.

*Evidence anchors:* [26,28]

### 11.0.6 4. Clinical utility

**Question.** Does using the measurement improve decisions or outcomes compared with current care?

**Design.** Randomized or pragmatic comparative study measuring downstream actions, benefits, harms, burden, equity, and cost.

**Success criterion.** Net benefit on patient-important outcomes with acceptable implementation burden.

**Failure or stop criterion.** Improved prediction without changed decisions or outcomes.

### 11.0.7 5. Longitudinal multimorbidity

**Question.** How does the domain interact with other conditions, treatments, and functional trajectories?

**Design.** Person-centered longitudinal cohort or trial with common data model, medication and intervention timelines, function, participation, and competing risks.

**Success criterion.** Models that support coordinated care rather than isolated disease optimization.

**Failure or stop criterion.** Single-disease endpoint worsens overall burden or conflicts with goals.

### 11.0.8 6. Implementation and governance

**Question.** Can effective interventions be delivered safely, equitably, affordably, and sustainably?

**Design.** Hybrid effectiveness-implementation studies, participatory design, workflow and workforce analysis, privacy/security review, and policy evaluation.

**Success criterion.** Sustained adoption with monitored outcomes and reduced inequity.

**Failure or stop criterion.** Pilot success depends on exceptional staffing or hidden externalized costs.

# Cross-volume integration

The health atlas is deliberately polyhierarchical. A phenomenon can have one primary home while also participating in multiple causal and functional systems. Cross-references below identify the most important companion volumes and the reason for the link.

| Volume | Companion topic                              | Relationship                                                                                                 | Use this link when...                                                                     |
|--------|----------------------------------------------|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| 00     | Atlas of Human Health                        | Master taxonomy, history of classification, and multi-axial architecture.                                    | locating the primary home of a concept or mapping classifications.                        |
| 13     | Measurement, Evidence, Data, and Informatics | Shared metrology, causal inference, validation, interoperability, and data governance.                       | designing a study, sensor, dataset, model, or evidence claim.                             |
| 14     | Interventions and Therapeutics               | Cross-domain pharmacology, procedures, behavioral interventions, rehabilitation, implementation, and safety. | moving from measurement to action.                                                        |
| 15     | Ethics, Law, Rights, Equity, and Governance  | Consent, privacy, fairness, rights, research oversight, and distribution of benefits and harms.              | people, identifiable data, invasive procedures, or consequential algorithms are involved. |
| 03C    | Nervous System Health                        | Sensory pathways, central processing, migraine, neurodegeneration and stroke.                                | organ tests are normal or deficits localize beyond the receptor.                          |
| 05     | Brain, Cognition, Emotion and Behavior       | Attention, prediction, language, hallucination, distress and learning shape perception.                      | symptom burden exceeds peripheral findings or development is central.                     |
| 10     | Function, Disability and Rehabilitation      | Communication, mobility, assistive technology and accessible environments determine participation.           | moving from impairment to real-world capability.                                          |
| 07     | Life Course                                  | Critical periods, education and aging change consequences and intervention timing.                           | screening or rehabilitation is age-dependent.                                             |
| 09     | Environmental and Occupational Health        | Noise, UV, chemicals, trauma and built-environment access are preventable determinants.                      | exposure control is part of treatment.                                                    |

## Cross-reference rule

Reference the primary home for the full mechanism or standard; restate only what is necessary for the local argument. When a claim spans domains, record the bridge explicitly rather than assigning the entire phenomenon to one specialty.

# Minimum data schema for a longitudinal domain profile

| Layer                     | Minimum fields                                                                                                                       | Critical caution                                                                      |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Identity and context      | Participant identifier, consent, demographics relevant to the question, setting, language, access, and social/environmental context. | Avoid collecting identity variables without purpose; preserve privacy and provenance. |
| Time                      | Absolute timestamps, timezone, age/life stage, episode boundaries, event markers, medication and exposure timing.                    | Clock drift and ambiguous time zones can invalidate multimodal inference.             |
| Raw signal / specimen     | Unprocessed samples, units, sensor channel, specimen type, collection protocol, calibration and storage state.                       | Derived scores without raw or quality data are difficult to audit.                    |
| Quality                   | Contact, motion, saturation, missingness, assay controls, operator notes, confidence, out-of-range flags.                            | Quality must be a first-class variable, not only an exclusion after analysis.         |
| Reference                 | Comparator method, instrument, version, observer protocol, laboratory traceability, adjudication.                                    | A weak reference caps the validity of the new system.                                 |
| Phenotype and function    | Symptoms, signs, diagnoses, severity, capability, participation, patient-reported outcomes, goals.                                   | Disease codes alone do not represent lived outcome.                                   |
| Intervention and response | Dose, route, device settings, behavior support, adherence, co-interventions, adverse events, response window.                        | Confounding by indication and treatment changes must be modeled.                      |
| Governance                | Protocol version, data-use agreement, access log, retention, deletion, model version, incident record.                               | Data stewardship is part of scientific validity.                                      |

# Glossary

| Term                       | Operational meaning in this volume                                                                                                                    |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Psychophysics              | Quantitative study of relationships between physical stimuli and perception/behavior.                                                                 |
| Contrast sensitivity       | Ability to detect differences in luminance or other signal contrast across spatial/temporal conditions.                                               |
| Sensorineural hearing loss | Hearing loss arising mainly from cochlea, auditory nerve or central pathways rather than sound conduction.                                            |
| Vestibulo-ocular reflex    | Reflex stabilizing gaze during head motion by generating compensatory eye movement.                                                                   |
| Allostasis                 | Adaptive regulation that changes physiological activity to meet anticipated or actual demands; chronic or repeated activation can carry cost.         |
| Biomarker                  | Defined characteristic measured as an indicator of normal biology, pathogenic process, exposure, or response; not automatically a surrogate endpoint. |
| Function                   | What a person can do in body functions, activities, and participation, interpreted in context.                                                        |
| Homeostasis                | Regulation of variables within viable ranges through feedback, feedforward, redundancy, and adaptation.                                               |
| Phenotype                  | Observable or measurable characteristics arising from biology, development, environment, behavior, context, and measurement conditions.               |
| Risk factor                | Characteristic associated with future outcome; causal status and modifiability must be established separately.                                        |
| Calibration                | Establishing the relationship between an instrument output and a reference quantity under specified conditions.                                       |
| Clinical validity          | Degree to which a measurement or model is associated with or predicts the clinical state or outcome it claims to represent.                           |
| Intended use               | The specific purpose, population, setting, user, input, and decision for which a device or model is designed.                                         |
| Reference method           | Comparator selected to estimate truth for validation; every reference has its own error and scope.                                                    |
| Reserve                    | Capacity to maintain function under challenge beyond resting baseline.                                                                                |
| Trajectory                 | Direction, rate, variability, and reversibility of change over time.                                                                                  |

# Selected references and current standards

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The bibliography emphasizes official standards, current guidelines, landmark primary work, and major syntheses. Inclusion does not imply that every recommendation is universal or that institutional endorsement removes uncertainty. Current online materials were checked through the stated evidence cutoff.

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# Version note

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Foundation edition generated August 2026. Evidence, guidance, and standards cutoff: 13 August 2026. The suite should be versioned when major classifications, guidelines, device standards, or decisive evidence materially change the map.